2. DOUBLE DES (2DES)

AIM

To implement the Double DES (2DES) algorithm using Python.

Algorithm

1. Start.
2. Import DES library.
3. Enter plaintext.
4. Enter two different 8-character keys.
5. Encrypt the plaintext using Key 1.
6. Encrypt the output again using Key 2.
7. Display the encrypted text.
8. Decrypt using Key 2.
9. Decrypt again using Key 1.
10. Display the original plaintext.
11. Stop.

Python Code

```
from Crypto.Cipher import DES

from Crypto.Util.Padding import pad, unpad

import base64

plaintext = input("Enter the Plain Text: ")

key1 = input("Enter First 8-character Key: ").encode()

key2 = input("Enter Second 8-character Key: ").encode()

cipher1 = DES.new(key1, DES.MODE_ECB)

cipher2 = DES.new(key2, DES.MODE_ECB)

step1 = cipher1.encrypt(pad(plaintext.encode(), DES.block_size))

encrypted = cipher2.encrypt(step1)

print("\nEncrypted Text :", base64.b64encode(encrypted).decode())

step2 = cipher2.decrypt(encrypted)

decrypted = unpad(cipher1.decrypt(step2), DES.block_size)

print("Decrypted Text :", decrypted.decode())
```

Sample Output

```
Enter the Plain Text: bhavya
Enter First 8-character Key: security
Enter Second 8-character Key: password

Encrypted Text : AW62q+H3Cs4=
Decrypted Text : bhavya
```

The result:

The above code is executed and the output is obtained successfully.